

C.W

04.10.2026

(*) $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$; find A^{-1} using character. eqn.

(2nd part)

Solution:

$$\lambda^3 - 14\lambda^2 + 46\lambda + 1 = 0$$

$$\Rightarrow A^3 - 14A^2 + 46A + 1 = 0$$

Multiply both sides A^{-1} , then we get,

$$A^{-1}A^3 - 14A^{-1}A^2 + 46A^{-1}A + A^{-1} = 0$$

$$\Rightarrow A^2 - 14A + 46 + A^{-1} = 0$$

$$\Rightarrow A^{-1} = -A^2 + 14A - 46$$

$$= - \begin{bmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 9 & 2 & 67 \end{bmatrix} + 14 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} - 46 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= - \begin{bmatrix} 8 & 12 & 33 \\ 15 & 29 & 45 \\ 9 & 2 & 67 \end{bmatrix} + \begin{bmatrix} 14 & 28 & 42 \\ 28 & 70 & 42 \\ 14 & 0 & 112 \end{bmatrix} - \begin{bmatrix} 46 & 0 & 0 \\ 0 & 46 & 0 \\ 0 & 0 & 46 \end{bmatrix}$$

$$= \begin{bmatrix} -8+14-46 & -12+28 & -33+42 \\ -15+28 & -29+70-46 & -45+42 \\ -9+14 & -2 & -67+112-46 \end{bmatrix}$$

$$= \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

(Ans)

* Obtain the eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix}$ and hence obtain an invertible matrix such that $P^{-1}AP$ is diagonal.

Solution: (1st Part)

We know,

$$|A - \lambda I| = 0$$

$$\Rightarrow \left| \begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\Rightarrow \left| \begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 4 \\ 9 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^2 - 36 = 0$$

$$\Rightarrow 1 - 2\lambda + \lambda^2 - 36 = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 35 = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 5\lambda - 35 = 0$$

$$\Rightarrow \lambda(\lambda-7) + 5(7-7) = 0$$

$$\Rightarrow (\lambda-7)(\lambda+5) = 0$$

$$\Rightarrow \lambda = -5, 7 \leftarrow \text{eigen value.}$$

Now,

$$[A - \lambda I]V = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{bmatrix} 1-\lambda & 4 \\ 9 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} x(1-\lambda) + 4y \\ 9x + y(1-\lambda) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

~~$$\Rightarrow \begin{bmatrix} x + 4y - x\lambda \\ 9x + y - y\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$~~

i.e.

$$(1-\lambda)x + 4y = 0$$

$$9x + (1-\lambda)y = 0$$

when, $x = -5$,

$$\left. \begin{aligned} (1+5)x + 4y &= 0 \\ 9x + (1+5)y &= 0 \end{aligned} \right\}$$

$$\text{or, } \left. \begin{aligned} 6x + 4y &= 0 \\ 9x + 6y &= 0 \end{aligned} \right\} \text{ or, } \left. \begin{aligned} 3x + 2y &= 0 \\ 3x + 2y &= 0 \end{aligned} \right\}$$

$$\text{or, } 3x + 2y = 0$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = v[\lambda I - A]$$

Let's y be a independent variable,

$$\text{so, } y = 3$$

$$\therefore 3x + 2 \cdot 3 = 0$$

$$\Rightarrow x = \frac{-6}{3} = -2$$

when,

eigen value is, $\lambda_1 = 5$, then eigen vector

$$\text{is } v_1 = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

~~when,~~
~~for $\lambda = 7$ we get~~

when, $\lambda = 7$

$$\left. \begin{aligned} (1-7)x + 4y &= 0 \\ 9x + (1-7)y &= 0 \end{aligned} \right\} \text{ or } \left. \begin{aligned} -6x + 4y &= 0 \\ 9x - 6y &= 0 \end{aligned} \right\}$$

$$\text{or } \left. \begin{aligned} 3x - 2y &= 0 \\ 3x - 2y &= 0 \end{aligned} \right\}$$

$$\text{or, } 3x - 2y = 0$$

Let, y be a independent variable, so, $y = 3$

$$\therefore 3x - 2 \cdot 3 = 0$$

$$\Rightarrow x = \frac{6}{3} = 2$$

when,

eigen value is $\lambda_2 = 7$, then, eigen vector

$$\text{is } v_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$